UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 25: Sparse Regression

Content:

1. Sparsity
2. Compressed Sensing
3. Restricted Isometry Property

1 Sparsity

At the end of the last lecture, we established the following result:

Theorem 1.1 Suppose $y = X\beta^* + \varepsilon$, where $X \in \mathbb{R}^{n\times d}$, ε has sub-Gaussian entries with variance proxy σ^2. Let $\max_i \left\|x^i\right\| \leq \sqrt{n}$, and $\hat{\beta} \in \arg\min_\beta \|y - X\beta\|_2$, then

$$\mathbb{E}_\varepsilon \left[\frac{1}{n}\|X\hat{\beta} - X\beta^*\|_2^2\right] \leq 4\sigma\sqrt{\frac{2\log(2d)}{n}} = O\left(\sigma\sqrt{\frac{\log d}{n}}\right)$$

Remark: Combining with ULS, the upper bound becomes

$$\min\left\{\sigma\sqrt{\frac{\log d}{n}}, \frac{r\sigma^2}{n}\right\}$$

where $r = \text{rank}(X^TX)$. A phase transition:

- If $r \gg \sqrt{n}$, then $r/n \gg \sqrt{\log d/n}$.
- If $r \ll \sqrt{n}$, then $r/n \ll \sqrt{\log d/n}$.

Transition between the slow and fast rates occur presicely at $r \sim \sqrt{n}$.

Definition 1.2 (Sparsity) We say β^* is k-sparse if $\|\beta^*\|_0 = |\{i : \beta_i \neq 0\}| \leq k$. In other words, β^* is k-sparse iff $\exists S \subset [d] = \{1, \cdots, d\}$, $|S| \leq k$ and $\beta_i^* = 0$ for any $i \in S$.

If we know β^* is k-sparse, denote $\mathcal{B}_0^d(k) = \left\{\beta \in \mathbb{R}^d : \|\beta\|_0 \leq k\right\}$. If $\beta^*, \hat{\beta} \in \mathcal{B}_0^d(k)$, we know $\hat{\beta} - \beta^* \in \mathcal{B}_0^d(2k)$. Given the basic inequality:

$$\|X\hat{\beta} - X\beta^*\|_2 \leq 2\left\langle \varepsilon, \frac{X\hat{\beta} - X\beta^*}{\|X\hat{\beta} - X\beta^*\|_2} \right\rangle \leq 2 \max_{\substack{S\subset[d] \\ |S|=2k}} \sup_{\omega:\text{supp}(\omega)=S} \left\langle \varepsilon, \frac{X\omega}{\|X\omega\|_2} \right\rangle$$

Let U_S be an orthonormal basis for the column space of X indexed by S, we can rewrite $X\omega / \|X\omega\|_2 = U_s a$, for some $a \in \mathcal{B}_2^{2k} = \{u \in \mathbb{R}^{2k} : \|u\|_2 = 1\}$

$$LHS \leq 2 \max_{\substack{S\subset[d] \\ |S|=2k}} \sup_{a\in B_2^{2k}} \langle \varepsilon, U_s a\rangle = 2 \max_{\substack{S\subset[d] \\ |S|=2k}} \sup_{a\in B_2^{2k}} \left\langle U_S^T\varepsilon, a\right\rangle = 2 \max_{\substack{S\subset[d] \\ |S|=2k}} \left\|U_S^T\varepsilon\right\|_2$$

By squaring both sides and taking expectation, we get

$$\begin{aligned}
\mathbb{E}\|X\hat{\beta} - X\beta^*\|_2^2 &\leq 4\mathbb{E}\left[\max_{S:|S|=2k} \left\|U_S^T\varepsilon\right\|_2^2\right] \\
&\leq 4\mathbb{E}\left[\max_{S:|S|=2k} \left(\left\|U_S^T\varepsilon\right\|_2^2 - \mathbb{E}[\left\|U_S^T\varepsilon\right\|_2^2]\right)\right] + 4\max_{S:|S|=2k} \mathbb{E}[\left\|U_S^T\varepsilon\right\|_2^2] \\
&\leq 4\mathbb{E}\left[\max_{S:|S|=2k} \left(\left\|U_S^T\varepsilon\right\|_2^2 - \mathbb{E}[\left\|U_S^T\varepsilon\right\|_2^2]\right)\right] + 8k\sigma^2
\end{aligned}$$

Since $\left\|U_S^T\varepsilon\right\|_2^2 - \mathbb{E}[\left\|U_S^T\varepsilon\right\|_2^2]$, is a centered collection of $\binom{d}{2k}$ sub-Exp r.v. with variance $O(\sigma^2)$, thus

$$\mathbb{E}\left[\max_{S:|S|=2k} \left(\left\|U_S^T\varepsilon\right\|_2^2 - \mathbb{E}[\left\|U_S^T\varepsilon\right\|_2^2]\right)\right] \leq C\sigma^2 \log\binom{d}{2k} \leq C\sigma^2 \cdot 2k \log\frac{ed}{2k} = O\left(\sigma^2 k \log\frac{d}{k}\right)$$

here we use the inequality $\binom{n}{k} \leq (en/k)^k$. Overall, once we have sparsity $\hat{\beta} \in \arg\min_{\beta\in\mathcal{B}_0^d(k)} \|y - X\beta\|_2$, then

$$\mathbb{E}\left[\frac{1}{n}\|X\hat{\beta} - X\beta^*\|_2^2\right] \leq O\left(\frac{\sigma^2}{n} k \log\frac{d}{k}\right)$$

2 Compressed Sensing

In classical regression problem, given $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, the goal is to recover β^* from the model $y = X\beta^* + W$. The classical setting is that the number of features p is fixed, with n grow, and the classical approach is to solve OLS: $\min \|y - X\beta\|_q$. The most common choice is $q = 2$, and as long as $q \geq 1$, it is a convex optimization problem.

In modern setting (high dimensional setting), we have $p \to \infty$, and $p \gg n$. The classical approach OLS would lead to overfitting when $p \gg n$. Under this setting, the most popular assumption is that β^* is sparse:

1. Noiseless case: $W = O$.
2. β^* is s-sparse ($\|\beta^*\|_0 \leq s$).

One possible approach is that

$$\min_{\beta \in \mathbb{R}^p} \|\beta\|_0 , \qquad \text{subject to } y = X\beta$$

In this approach, $\|\beta\|_0$ is non-convex, and this program is NP-hard. If we replace $\|\cdot\|_0$ norm with $\|\cdot\|_1$, the problem becomes

$$\min_{\beta \in \mathbb{R}^p} \|\beta\|_1 , \qquad \text{subject to } y = X\beta$$

Now this becomes a convex optimization problem, and under some assumptions on X (which will be satisfied for sub-Gaussian X), we would have $\hat{\beta} = \beta^*$.

Roadmap:

1. If X satisfies **Restricted Isometry Property** (RIP), then β^* is the unique solution to the LP above.
2. If X has i.i.d. sub-Gaussian entries, then X satisfies RIP with high probability if $n \gg Cs \log(p/s)$ for some constant $C > 0$.

3 Restricted Isometry Property

Definition 3.1 (Restricted Isometry Property) The matrix $X \in \mathbb{R}^{n \times p}$ satisfies the RIP of order s and parameter $\delta \in (0, 1)$, if for every s-sparse $\beta \in \mathbb{R}^p$

$$(1 - \delta) \|\beta\|_2^2 \leq \|X\beta\|_2^2 \leq (1 + \delta) \|\beta\|_2^2$$

the inequality above holds. We can also say that X is (s, δ)-RIP.

Implications: X is (s, δ)-RIP iff for every $S \subset [p]$ with $|S| = s$, and every $\beta \in \mathbb{R}^s$

$$\begin{aligned} \left| \|X_S\beta\|_2^2 - \|\beta\|_2^2 \right| &\leq \delta \|\beta\|_2^2 \\ \Leftrightarrow |\langle X_S\beta, X_S\beta\rangle - \langle\beta, \beta\rangle| &\leq \delta \|\beta\|_2^2 \\ \Leftrightarrow |\left\langle X_S^T X_S\beta, \beta\right\rangle - \langle\beta, \beta\rangle| &\leq \delta \|\beta\|_2^2 \\ \Leftrightarrow |\left\langle (X_S^T X_S - I_S)\beta, \beta\right\rangle| &\leq \delta \|\beta\|_2^2 \end{aligned}$$

so X is (s, δ)-RIP iff $\|X_S^T X_S - I_S\|_{2\to 2} \leq \delta$.